# IPv4 Addressing and Subnetting Assignment Report

## 1. IPv4 vs IPv6

| IPv4 | IPv6 |
|---|---|
| 32-bit address | 128-bit address |
| Dotted decimal notation | Hexadecimal notation |
| Limited address space | Extremely large address space |
| Supports NAT widely | Usually no NAT required |

## 2. Home WiFi Static IP Assignment

| Device | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|
| Laptop | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 |
| Phone | 192.168.10.11 | 255.255.255.0 | 192.168.10.1 |
| Smart TV | 192.168.10.12 | 255.255.255.0 | 192.168.10.1 |
| Printer | 192.168.10.13 | 255.255.255.0 | 192.168.10.1 |

## 3. /27 Subnet Calculation

A /27 mask leaves 5 host bits. Hosts per subnet = 2^5 - 2 = 30 usable hosts. Borrowing 3 bits from a /24 network gives 2^3 = 8 subnets.

```
Subnet 1: 192.168.1.0 - 192.168.1.31
Usable Hosts: 192.168.1.1 - 192.168.1.30

Subnet 2: 192.168.1.32 - 192.168.1.63
Usable Hosts: 192.168.1.33 - 192.168.1.62

Subnet 3: 192.168.1.64 - 192.168.1.95
Usable Hosts: 192.168.1.65 - 192.168.1.94
```

## 4. Address Analysis for 10.0.5.17 /24

```
Subnet Mask: 255.255.255.0
CIDR: /24
Network Address: 10.0.5.0
Broadcast Address: 10.0.5.255
Valid Hosts: 10.0.5.1 - 10.0.5.254
```

## 5. Verification Using Online Subnet Calculator

The manual calculation matches the expected output of a subnet calculator for network address, broadcast address, and valid host range. A real submission should include an actual screenshot taken from the subnet calculator website.